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Current Opinion in Genetics & Development

Paternal genome elimination: patterns and mechanisms of drive and silencing

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In thousands of arthropod species, males inherit, but subsequently eliminate the entire haploid genome of their father. However, why this peculiar reproductive strategy evolved repeatedly across diverse species and what mechanisms are involved in paternal genome elimination (PGE) remains largely unknown. In this review, we summarize what we know about the patterns of paternal chromosome elimination during various stages of development in the diverse taxa that have been studied. We also discuss some other unusual features often associated with PGE, such as the transcriptional silencing of paternally derived chromosomes in males and sex determination through the early embryonic elimination of X chromosomes. Little is known about the molecular mechanisms underlying the parent-of-origin-dependent chromosome elimination and silencing under PGE, but we discuss the insight of several studies that are pioneering this work and highlight directions for future research.

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Introduction

The ability of organisms to pass on heritable material to future generations is perhaps the single most important trait to define living beings. In most animals, inheritance is Mendelian, with parents contributing an equal share of genetic material to offspring and each offspring passing on an equal mix of their two parental haplotypes. However, unequal inheritance of certain chromosomes, or even entire haploid genomes, has evolved repeatedly [1,2]. Perhaps, the most extreme example is found in species with paternal genome elimination (PGE), a reproductive strategy where males inherit genetic material from both their parents, but only ever pass on their mother's gene to their offspring. This unequal transmission results from meiotic drive of the maternal genome, as well as in some cases, the earlier elimination of paternal chromosomes. PGE evolved at least 6 times independently across arthropods [3] and is a phenomenon that is not transient, but instead an integral part of the life cycle of large numbers of species within each clade.

Despite PGE's widespread occurrence, we still know little about why and how it evolved. There is theory showing PGE can evolve as a result of conflict between parents [3–6], with mothers increasing their genetic contribution to future generations by evolving mechanisms to ensure that their sons preferentially pass on their genome. They also show that transitions to PGE require changes in sex determination and that conflict between maternal and paternally derived genomes (from now on referred to as maternal genome (MG) and paternal genome (PG) might select for the silencing or early loss of the PG. However, to validate these models, it is crucial to understand the molecular mechanisms responsible for 1) chromosome elimination and transcriptional silencing and 2) the mechanisms of sex determination. Here, we provide a summary of what is known from the cytogenetic literature and more recent genomic work. We finish by highlighting gaps in our knowledge and avenues for future research.

Paternal genome elimination across taxa

PGE is found across six clades of insects, springtails and mites (Table 1, Figures 1 and 2) [1,3]. While all PGE males inherit, but subsequently fail to pass on their PG, the timing and mechanism of loss differ; both the number of chromosomes eliminated (sex chromosomes or whole genome) and the time of elimination (during embryogenesis or spermatogenesis) vary. In PGE species with sex chromosomes (sciarid flies, gall midges and symphyleonian springtails), elimination of the paternal X chromosomes occurs early in development, followed by elimination of the entire PG during male meiosis. In other PGE organisms, the PG is eliminated in its entirety, either during embryogenesis or during spermatogenesis and in some cases also transcriptionally silenced. In the next section, we will briefly review patterns and mechanisms of elimination.

Table 1

Patterns of PGE across taxa.

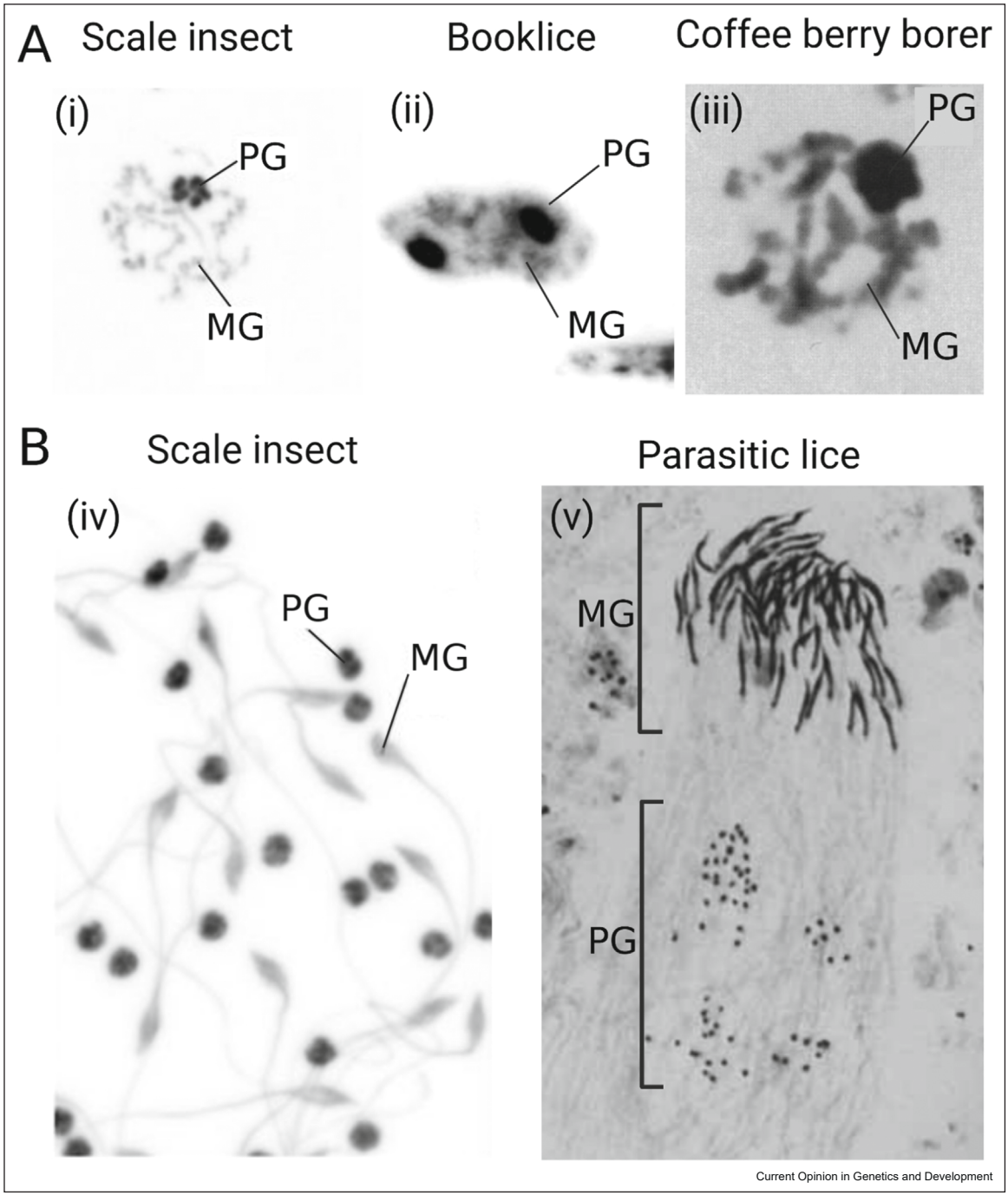
Class	Order	Clade	Number of species[a]	Chromosome eliminated	Type of PGE	Male soma	Sex determination karyotype[b]	Meiosis	Timing of elimination
Acari	Parasitiformes	Phtyoseiidae (Phtyoseiid mites)	2000 (10)	Whole paternal genome	Embryonic	Haploid	Haplodiploid	1 division	Early embryogenesis 24–48 hours post fertilization
Collembola	Symphypleona	Symphypleona (globular springtails)	1180 (6)	Paternal X and whole genome	Embryonic and germline	Diploid with X elimination	X_1X_200[c]	2 divisions	Early embryonic development, and first meiotic division
Insecta	Hemiptera	Neococcidae (scale insects, e.g. mealybugs)	4000 (270)	Whole paternal genome	Germline	Diploid with paternal genome silencing	No identifiable sex chromosomes	1 or 2 divisions	During second meiotic division (reductional), just before meiosis, and during second division if some chromosomes remain
Insecta	Hemiptera	Diaspididae (armored-scale insects, nested within the Neococcidae)	2650 (139)	Whole paternal genome	Embryonic	Haploid	No identifiable sex chromosomes	1 division	Early embryogenesis 6th cleavage Blastula stage
Insecta	Diptera	Sciaridae (fungus gnats)	2300 (10)	Paternal X and whole genome	Embryonic and germline	Diploid with X elimination	X0	2 divisions	Early embryogenesis for Xp, first meiotic division for autosomes
Insecta	Diptera	Cecidomyiidae (gall midges)	6168 (22)	Paternal X and whole genome	Embryonic and germline	Diploid with X elimination	X_1X_200[c]	2 divisions	Early embryogenesis for Xp, first meiotic division for autosomes
Insecta	Coleoptera	Hypothenemus (coffee berry borer beetles)	190 (2[d])	Whole paternal genome	Germline	Diploid with paternal genome silencing	No identifiable sex chromosomes	2 divisions	Eliminated during first division
Insecta	Psocodea	Psocodea (booklice)	? (1)	Whole paternal genome	Germline	Diploid with paternal genome silencing	No identifiable sex chromosomes	2 divisions	Eliminated post meiosis (or during 2nd inverted meiosis)
Insecta	Phthiraptera	Phthiraptera (parasitic lice)	? (3)	Whole paternal genome	Germline	Diploid	No identifiable sex chromosomes	2 divisions	Eliminated post meiosis (or during 2nd inverted meiosis)

Table 1 (*continued*)

Class	Order	Clade	Number of species[a]	Chromosome eliminated	Type of PGE	Male soma	Sex determination karyotype[b]	Meiosis	Timing of elimination
Parasitiformes			1						
		ACARI							
Symphypleona			1						
		COLLEMBOLA							
Hemiptera			1						
Phtiraptera			1						
Psocoptera			1						
Coleoptera			1						
Diptera			2						
		INSECTA							

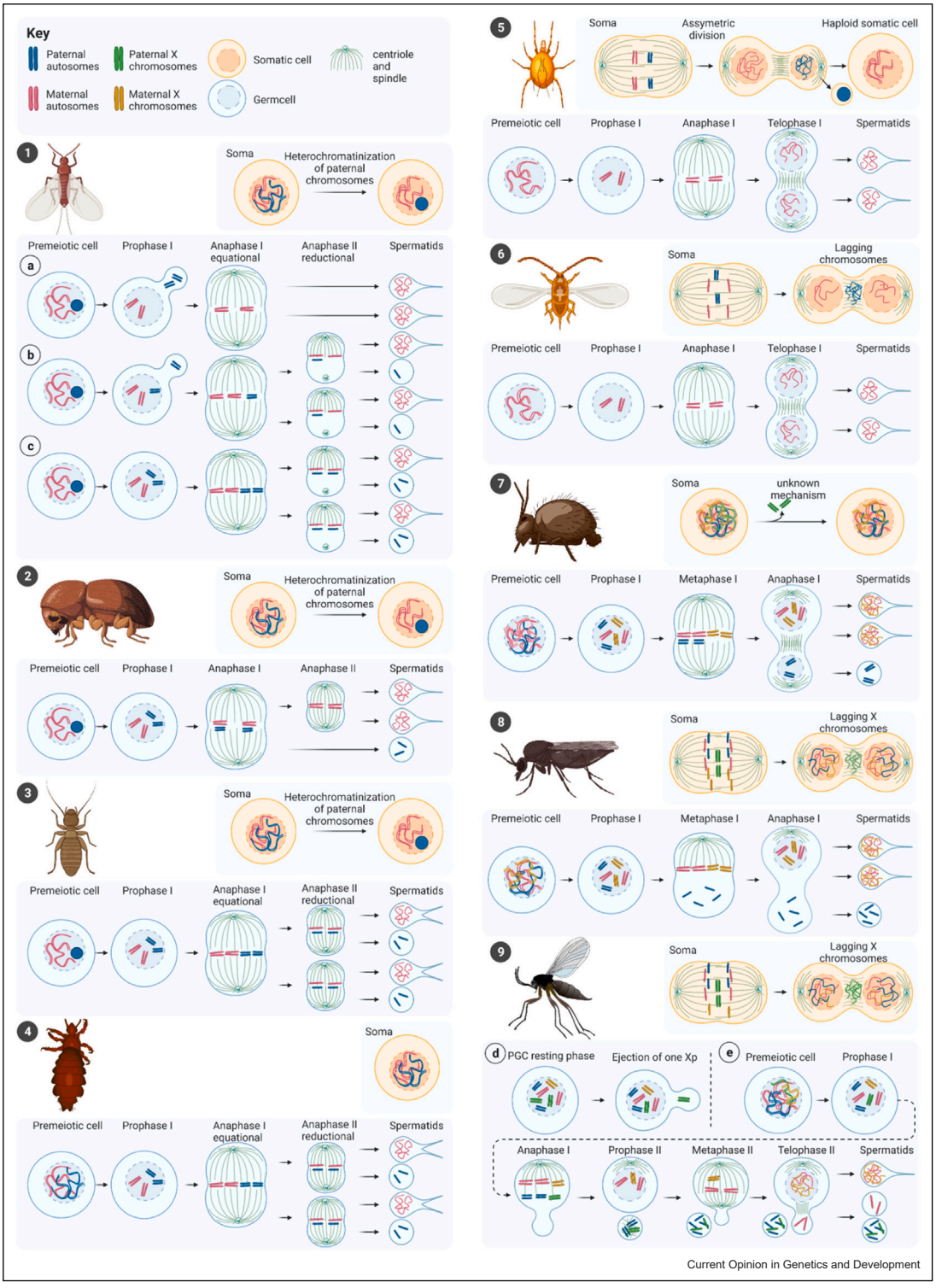
[a] Number of species column: estimated number of species in one clade assuming PGE is conserved; numbers of species with evidence of PGE are in parentheses.
[b] Sex determination column: sex determination refers to the karyotype differences between adult males and females in the PGE lineage.
[c] X_1X_200 represents heterogametic sex determination of males with two X chromosomes (females are homogametic $X_1X_2X_1X_2$).
[d] *Hypothenemus obscurus* probably has PGE although not fully confirmed [56].

Figure 1



Similar phenotypes in different PGE species. **(a)** DNA staining showing heterochromatinization of the PG and euchromatic MG, in germline cells from male testis at early prophase I in scale insect (*P. viburni*) (i), cells from male abdominal tissue in booklice (*Liposcelis* sp.) (ii), male neuroblast in prophase from the coffee berry borer (*H. hampei*) (iii). **(b)** DNA staining of sperm maturation in scale insect (*P. viburni*) (iv) and parasitic lice (*Haematopinus suis*) (v) showing round highly condensed nuclei carrying the PG. These are being eliminated from the bundle of elongating spermatids carrying the MG.
Images adapted from [22,32,44,50]. Copyright (1995) National Academy of Sciences. Copyright (1993) Nature Springer.

Figure 2



Overview of PGE during male meiosis and in somatic cells in (1) scale insects (Coccoidea, except armored scales), (2) coffee berry borer *Hypothenemus hampei*, (3) *Liposcelis* booklice, (4) parasitic lice, (5) predatory mites (Phytoseiidae), (6) armored-scale insects (Diaspididae), (7) globular springtails (Symphypleona), (8) gall midges (Cecidomyiidae), (9) dark-winged fungus gnats (Sciaridae). In somatic cells, the entire PG is heterochromatinized (1–3), eliminated via an asymmetric division (5), or via chromosomes lagging on the metaphase plate (6), while in species with sex chromosomes, the two paternal X chromosomes are eliminated during mitotic division in early embryogenesis (7–9) via chromosome lagging (8–9) or an unknown mechanism (7). The soma is therefore either truly (5–6) or functionally haploid, or diploid for the entire genome (4) or just the autosomes

(7–9). In the germline, the PG is eliminated through a highly modified meiosis (1). In scale insect, either all paternal chromosomes **(a)** or a subset **(b)**, or none **(c)**, are eliminated during early prophase I. If no paternal chromosomes remain after prophase I **(a)**, then maternal chromosomes engage in a single meiotic division. If any paternal chromosomes remain after prophase I, then in those systems with an inverted meiosis, the sister chromatids are separated first (equational division), and then in the reductional division, the MG and PG are separated through a monopolar spindle that attaches to the maternal genome only. Spermatids carrying the MG elongate, while nuclei with the PG degrade. (2) In coffee berry borer, PG elimination occurs during meiosis I, one nucleus containing the PG degrades, while the MG engages in meiosis II to produce two identical spermatids. (3–4) In booklice and parasitic lice, meiosis is probably inverted and PGE occurs during the second reductional division, producing half biflagellate spermatids and half degrading nuclei. (5–6) In mites and armored-scale insects, PGE occurs during early development before germline/soma differentiation, thus, somatic and germ cells are haploid, the latter engaging in a unique meiotic division. (7–8) In springtails and gall midges, PGE occurs during meiosis I through an asymmetric division where the two maternal X chromosomes and autosomes are separated from the paternal autosomes, while the second meiotic division is canonical. (8) In gall midges, a monopolar spindle is present during meiosis I that attaches to maternal chromosomes only. (9) In fungus gnats, two separate elimination events occur in the germline: **(d)** during early embryonic development, one paternal X (Xp) is ejected from the nuclei and degraded; **(e)** during meiosis, maternal and paternal chromosome attach to a monopolar spindle, but while maternal chromosomes are pulled toward the cell pole, paternal chromosomes move away from the single pole and are extruded into a cytoplasmic bud lined up with microtubules; in meiosis II, a bipolar spindle is formed to separate maternal autosomes, but there is non-disjunction of the X dyad. The double-X cell constitutes the only viable product of male meiosis; the nullo-X cell is eliminated as a cytoplasmic bud. Thus, both sister chromatids of the maternal X end up in the spermatid. Figure created with BioRender.

Embryonic elimination

In predatory mites (Phtyoseiidae) and armored-scale insects (Diaspididae), elimination of the entire PG occurs during early development, through condensation and exclusion of the PG during an asymmetrical division in mites [7], or through non-disjunction of paternal chromosomes during anaphase in armored-scale insects [8]. In both, meiosis is a simple mitotic-like division that produces two genetically identical haploid spermatids.

In species with sex chromosomes (springtails, fungus gnats and gall midge, Table 1 and Figure 2), the elimination of paternal X chromosomes occurs early during male embryonic development, followed by the elimination of the remaining PG during male meiosis. Embryos of both sexes begin with equal numbers of Xs (four Xs for springtail and gall midges, and three for sciarid flies) [9–14]. In male embryos, paternal X chromosomes (Xp) fail to separate in anaphase and are eliminated, while maternal Xs (Xm) divide normally. During spermatogenesis, the PG is eliminated during the first meiotic division in all taxa [9,12,15–17]. Moreover, in sciarid flies, two additional events occur, in primordial germ cells, one Xp is extruded during a long nondividing resting phase [18]; subsequently during meiosis II, the sister chromatids of the Xm remain attached and constitute the two Xp after fertilization [19–21].

Germline elimination

In other taxa, elimination of the entire PG is delayed until spermatogenesis. In the coffee berry borer *Hypothenemus hampei*, paternal chromosomes do not pair during meiosis and degenerate *in situ*, while the maternal chromosomes engage in an equational division, forming two haploid spermatids [22]. In most scale insects, including mealybugs, the PG is eliminated during a highly modified meiotic sequence [23–26]: the usual order in which meiosis occurs is inverted, with the homologs separating during the second rather than the first division and involving a monopolar spindle that attaches to maternal chromosomes only [27]. Just those spermatids that receive the maternal chromosomes elongate into mature sperm, while spermatids carrying the paternal chromosomes degenerate (Figure 1) [28,29]. In a few-scale insect taxa, the elimination of several, and in some cases all paternal chromosomes, can also occur just before meiosis [23,30]. Meiosis in parasitic lice is highly modified with achiasmatic meiosis, followed by several rounds of mitotic division at the end of which an asymmetrical division produces functional spermatids carrying the MG, and degrading nuclei carrying the PG [31,32]. This last division might be the second meiotic division, which implies that the sequence of meiosis is inverted, as in scale insects, although this remains to be confirmed.

Paternal genome expression and ploidy

Different PGE species vary in the timing of PG elimination and whether the PG becomes transcriptionally inactive or not (Table 1). In species with whole PG elimination during embryogenesis, males develop as true haploids [7,30], while in those that eliminate X chromosomes, males will be haploid for their X chromosome but diploid for their autosomes. Mealybugs, coffee berry borers, and booklice remain diploid, however, paternal chromosomes become highly condensed in early development and cluster at the periphery of the nucleus to become transcriptionally silenced (Figure 1) [22,30,33]. Males are therefore functionally haploid [34]. Cytological and chromatin immunoprecipitation analyses in mealybugs and booklice reveal that paternal chromosomes are enriched for the constitutive heterochromatin mark H3K9me2/3 [33,35,36], and in mealybugs also for the facultative heterochromatin mark H3K27me3 and DNA methylation, all of which are associated with gene repression [35,37]. There are some PGE species where males remain fully diploid: the human body and head lice *P. humanus* lack PG heterochromatinization in somatic and germline cells [31] and a recent gene expression study shows diploid expression in males [38].

Mechanisms of imprinting and drive

A crucial aspect of PGE is that chromosome behavior depends on the parent-of-origin (imprinting) and therefore requires a mechanism for the recognition of paternally derived chromosomes. Imprinting in mammals is often regulated by DNA methylation, which has also been found in mealybugs (36) and the fungus gnat *Bradysia coprophila* (cite Urban et al.), even though dipteran flies lack DNA methyltransferases 1 and 3. *B. coprophila* also has 6 mA as well as the 6-mA methylase (DAMT-1) and demethylase (dMAD) [39]. There are extensive differences in the distribution of histone modifications between the MG and PG in many taxa. Some of these are likely related to PG silencing, such as H3K9me2/3 (see above), and might be unrelated to parental recognition. In sciarids, paternal chromosomes become decorated with H3K4me2 before elimination and phosphorylated for histone H3 during meiosis I (correlated with chromosome condensation and kinetochore function) [40,41]. Again though, it is unclear if these marks are involved in imprinting or elimination.

Another common feature under PGE is the preferential transmission of the MG during meiosis. At this point, nothing is known about the molecular mechanisms underlying this process, but they might be similar to other chromosomal drive systems — where one chromosome gains transmission over another — notably those in Diptera (reviewed in Ref. [42]). In the *Drosophila* SD system, spermatids lacking the driver display post-meiotic defects and fail to assemble proper spermatid chromatin, leading to their degradation [43]. This resembles mealybug and louse spermiogenesis, where spermatids containing paternally derived chromosomes degrade [32,44,45]. In sex chromosome drive systems (e.g. *D. simulans* Paris sex ratio), Y chromosomes lack centromeric activity or fail to separate sister chromatids in anaphase II [42,46]. These defects are reminiscent of what we see during X elimination under PGE.

Sex determination

Sex determination under PGE remains poorly understood. PGE evolved in clades with XY or X0 males. However, chromosomal sex determination is incompatible with PGE: X0/XY PGE males will always pass on their (maternal) X and therefore all their offspring would be female. While this female bias might have helped PGE spread, as it can be favorable under certain mating systems, it means that for PGE to fix in a population, a new sex determination mechanism is required [3]. One sex determination mechanism that has evolved repeatedly in PGE taxa is the elimination of one or multiple Xs from embryos that initially have an identical number of X chromosomes, but that become either XO (male) or XX (female) after elimination (Figure 2). As elimination occurs before the onset of embryonic gene expression, it is likely under control of maternal factors. In some sciarids and gall midges, this maternal factor is genetically determined, with females producing either all male or all female offspring, depending on the presence of an inversion (either X-linked or autosomal) [47].

Far less is known about sex determination in species without X chromosomes, where there is no genetic difference between males and females at any point during development. There is a recent report of a possible male-specific chromosome in *H. hampei* (coffee berry borer) [48], however, it is unclear if this chromosome is involved in sex determination. In clades where the PG is silenced early in development, the resulting haploidization might play a role in triggering male development, although that still leaves the question as to what causes paternal silencing in embryos destined to become males.

Long-term persistence or loss of paternal genome elimination

Most chromosomal drive systems tend to be transient, polymorphic in populations, and rarely persist over long evolutionary time [1]. PGE appears to be a clear exception. So far, we find that drive persists in all clades it has evolved in Ref. [23], the oldest estimated to be ≈400MYA [49]. There are however cases where particular chromosomes appear able to escape elimination. One example is a B chromosome in the mealybug *Pseudococcus viburni*, which is present in > 70–100% of all sperms, even when paternally derived [44]. Another example are two large germline-restricted chromosomes found in sciarids, which escape PGE during meiosis [50]. Finally, in body lice, several studies have found incomplete PG elimination, where only some paternal alleles were excluded, however without a consistent pattern, suggesting it is not a particular chromosome that escapes [51,52]. It is currently not clear how incomplete PGE occurs and what distinguishes driving and non-driving chromosomes. Better understanding this could be an important strategy to study the mechanisms of elimination.

Future directions

PGE has evolved repeatedly across arthropods and shows remarkable convergent evolution, with associated features, including PG silencing and X elimination, evolving in multiple clades (Figure 1). While there is a wealth of cytogenetic work characterizing patterns across species, we have little insight into the underlying molecular mechanisms. Until now, it was hard to develop the tools and resources required for mechanistic analyses in species with PGE. However, technological advances now make this feasible. In particular, the sciarid *B. coprophila* is proving a promising system, with a high-quality reference genome now available [39,47] and a

transformation system developed to allow for experimental investigation of gene function [53]. Besides mechanistic aspects, there are many other outstanding questions: one aspect that has barely been explored is how the unusual patterns of chromosome expression and transmission will affect molecular evolution, including rates of adaptation, diversification, sexually antagonistic selection, and the accumulation of transposable elements [54,55]. Another is how PGE interacts with a species' ecology, many PGE clades have particular ecological niches, often as ectoparasites of plants or animals, but it remains unclear why. To conclude, the study of PGE could provide insights into a wide range of topics in cell biology, genetics, and evolution. However, it will be crucial to go beyond descriptive science to really harbor its potential.

Data Availability

No data were used for the research described in the article.

Declaration of Competing Interest

None.

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